

Statistics Guide — algorithms and methods behind Tab 3

This document explains **how** each column in the Tab 3 “Generate Statistics” CSV is computed — the specific algorithm, library, and formula behind it — not just what the number means. For a plain-English, biology-oriented explanation of each column instead, see [GUIDE.md, Section 11](#). This document is the companion reference for anyone extending `_statistics.py`, auditing a result, or writing up the method (e.g. a Methods section).

All of this lives in one module, `napari_zf_microglia_ai/_statistics.py`, entered through a single function, `compute_stats()`.

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1. Pipeline architecture

`compute_stats(labels, scale_zyx, image=None, region_lines=None, region_names=None, backend_config=None)` runs in three phases, each chosen to keep the expensive parts vectorised or parallelised rather than looping over labels one at a time in pure Python:

Phase	What	How
1	Batch regionprops (volume, centroid, bbox, inertia tensor, solidity, extent) for every label in one call	cuCIM (GPU) if available, else scikit-image (CPU)
2a	Per-label marching-cubes surface area + skeleton/branch analysis	ThreadPoolExecutor, one worker per label, cropped to that label's bounding box
2b	Per-label intensity statistics (<i>only if an Image layer was supplied</i>)	Same thread-pool pattern as 2a
3	Assemble the DataFrame, then compute statistics that need <i>all</i> labels at once (nearest-neighbour, brain region), then generate descriptions	Vectorised NumPy/SciPy, then one description call per row

Why cropping to the bounding box matters (2a/2b): marching cubes and skeletonization are run on a small 3D crop (`labels[z0:z1, y0:y1, x0:x1] == 1b1`), not the full volume — this is what makes per-label operations on a multi-megavoxel image tractable at all. The crop bounds come directly from Phase 1's regionprops bbox output, so Phase 1 must run first.

GPU backend detection (`_detect_stats_backend()`): tries importing `cupy + cucim.skimage.measure`, then runs a 4×4×4 smoke-test `regionprops_table` call to confirm the CUDA JIT actually compiles and runs (not just that the import succeeded) before committing to the GPU path for the whole run. Falls back to CPU

(`skimage.measure.regionprops_table`) on any failure — including cuCIM being unavailable on Windows, since it has no native Windows wheels (see README/GUIDE’s Windows install notes).

Thread count: `_N_THREADS = max(1, os.cpu_count() // 2)` — half the logical cores, leaving headroom for the GPU-bound phase and the main thread rather than saturating every core.

2. Phase 1 — Batch regionprops

`_batch_regionprops(labels)` calls `regionprops_table` (cuCIM or `skimage` — same API, same property names) once, requesting:

```
_RPROPS = [  
    "label", "area", "centroid", "bbox",  
    "inertia_tensor", "inertia_tensor_eigvals",  
    "axis_major_length", "axis_minor_length",  
    "solidity", "extent",  
]
```

This single call is where most of the raw geometry comes from:

Column(s)	Source property	Notes
volume_vox	area	scikit-image calls the 3D voxel count “area” for historical (2D-first) API reasons
centroid_z/y/x_vox, centroid_z/y/x_um	centroid	voxel centroid × <code>scale_zyx</code> for the μm columns
bbox_z0/y0/x0_vox (start, inclusive), bbox_z1/y1/x1_vox (end, exclusive)	bbox	standard NumPy half-open slicing convention — <code>labels[z0:z1, y0:y1, x0:x1]</code> is exactly the label’s bounding crop
bbox_dz/dy/dx_um	bbox (end – start) × <code>scale_zyx</code>	physical size of the bounding box, not the label’s own extent (a diagonal or branched cell can have a much larger

Column(s)	Source property	Notes
		bbox than its actual volume would suggest — see extent below)
solidity	solidity directly	volume / convex_hull_volume — the library computes the 3D convex hull internally and divides
extent	extent directly	volume / bounding_box_volume
axis_major_length, axis_minor_length	used below	pixel-space ellipsoid-fit lengths, not yet in μm — see §3 for why these need care under anisotropic voxels
$\text{volume_um}^3 = \text{volume_vox} \times \text{sz} \times \text{sy} \times \text{sx}$ (vox_vol = the physical volume of one voxel, computed once from scale_zyx).		

3. Derived shape metrics from the inertia tensor

This is the part of the pipeline most worth reading carefully if you're citing these numbers, because it involves a real approximation under anisotropic voxel scaling ($Z = 1.0 \mu\text{m}/\text{vox}$ vs. $Y = X = 0.174 \mu\text{m}/\text{vox}$ in this project's typical data).

3.1 principal_axis_dir — which axis is the cell elongated along

The 3×3 inertia tensor for every label (from Phase 1) is stacked into an $(N, 3, 3)$ array and eigendecomposed in one batched call, `np.linalg.eigh(IT)`. `eigh` returns eigenvalues in **ascending** order — physically, the eigenvector paired with the **smallest** eigenvalue of the

inertia tensor is the object's axis of elongation (mass concentrated close to that axis contributes little to inertia about it; mass spread far from an axis contributes a lot). So:

```
longest_vecs = eigvecs[:, :, 0] # smallest-eigenvalue
eigenvector = major axis direction
axis_dirs = ["Z", "Y", "X"][argmax(|component|)] # per label,
whichever axis the eigenvector points closest to
```

principal_axis_dir is therefore the axis (Z, Y, or X) that the true 3D major axis most nearly aligns with — not a continuous angle, just the dominant of the three.

3.2 axis1_um, axis2_um, axis3_um — the anisotropy caveat

axis_major_length/axis_minor_length from regionprops are computed in **voxel units**, from an ellipsoid fit that implicitly assumes isotropic voxels. Converting to μm correctly requires knowing *which physical axis* the length actually runs along — a length “along Z” and a length “along X” need different scale factors (1.0 vs. 0.174 $\mu\text{m}/\text{vox}$ here) to become physically correct.

The code handles this only for the *major* axis, by construction:

```
axis1_scales = [axis_scale_map[d] for d in axis_dirs]
# pick sz, sy, or sx per-label, matching
# principal_axis_dir
a1_um = axis_major_length * axis1_scales
a3_um = axis_minor_length * scale_mean #
scale_mean = (sz+sy+sx)/3 — an approximation
a2_um = (a1_um + a3_um) / 2.0 # NOT
independently measured — see below
```

Three things worth knowing if you rely on these numbers:

- **axis1_um is scaled correctly** (per-label, using whichever physical axis it actually points along).
- **axis3_um uses the mean of the three axis scales**, not the true axis it points along, because regionprops' minor-axis direction isn't tracked per-label the way the major axis is here. This is an approximation, more accurate the closer $sz/sy/sx$ are to each other (it is *not* a negligible approximation for this project's typical ~5.7:1 Z:XY anisotropy on an axis pointing mostly along Z or mostly along XY).
- **axis2_um is not measured at all** — it's simply $(\text{axis1_um} + \text{axis3_um}) / 2$, a placeholder for the true middle principal axis (which would require a second eigenvector-aligned length

measurement that regionprops doesn't provide directly). Treat `axis2_um` as illustrative, not a real geometric measurement.

3.3 elongation, `eq_diam_um`

- `elongation` = `axis1_um / axis3_um` (guarded against divide-by-zero with a $1e-10$ floor). 1.0 = sphere; higher = more elongated.
 - `eq_diam_um` = $(6 \cdot \text{volume_um}^3 / \pi)^{1/3}$ — the diameter of a sphere with the same volume as the label. Shape-independent; two cells with identical volume have the same `eq_diam_um` regardless of how elongated or branched either one is.
-

4. Phase 2a — Surface area (marching cubes) and sphericity

`_surface_area(binary, scale_zyx):`

1. The label's cropped binary mask is **padded by 1 voxel on every side** (`np.pad(binary, 1)`) before meshing. This is necessary because marching cubes needs a zero-boundary to close the mesh — without padding, a label touching its own crop's edge would produce an open (non-manifold) surface and an undercounted area.
2. `skimage.measure.marching_cubes(padded, level=0.5, spacing=scale_zyx)` extracts an isosurface at the 0.5 threshold (the boundary between labeled and unlabeled voxels), directly in physical units via the `spacing` parameter (so no separate unit conversion is needed afterward).
3. `skimage.measure.mesh_surface_area(verts, faces)` sums the area of every triangle in the resulting mesh.
4. Returns `0.0` on any failure (e.g. a 1-voxel object with no meaningful surface) rather than raising, so one bad label doesn't abort the whole run.

Sphericity (computed in the main assembly loop, not inside `_surface_area`):

$$\text{sphericity} = \pi^{1/3} \cdot (6V)^{2/3} / A \quad (V = \text{volume_um}^3, A = \text{surface_area_um}^2)$$

This is the classical Wadell (1935) sphericity — the surface area a perfect sphere of the same volume *would* have, divided by the actual surface area. It's clamped to a maximum of 1.0 (mesh discretisation

noise can occasionally push the raw ratio fractionally above 1.0 for near-spherical labels; a physical sphericity cannot exceed 1.0 by definition).

`surface_to_volume_ratio = surface_area_um2 / volume_um3` — a simpler, non-normalised alternative to sphericity; unlike sphericity it keeps growing without bound as branching increases, rather than saturating.

5. Phase 2a — Skeleton and branching statistics

`_skeleton_stats(binary, scale_zyx)`, using `skimage.morphology.skeletonize` + the `skan` package:

1. `skeletonize(binary)` — thins the 3D binary mask to a topological skeleton (1-voxel-wide medial curve), preserving connectivity and branch structure.
2. `skan.Skeleton(skeleton, spacing=scale_zyx, source_image=binary)` builds a graph representation, with edges already in physical units via `spacing`.
3. `skan.summarize(sk, separator="-")` produces one row per skeleton **branch** (an edge between two graph nodes — a branch point or an endpoint), with columns including `branch-type`, `euclidean-distance`, and `branch-distance` (path length along the branch).

From that branch table:

Column	Computation
<code>n_branches</code>	<code>len(branch_table)</code> — one row per branch
<code>n_endpoints</code>	count of rows where <code>branch-type == 1</code> (skan's code for a branch ending in a free tip, as opposed to a branch connecting two junctions)
<code>mean_branch_len_um</code>	mean of <code>euclidean-distance</code> (straight-line branch endpoint-to-endpoint distance, not path length) across all branches
<code>max_branch_len_um</code>	

Column	Computation
	max of the same euclidean-distance column
branch_tortuosity	mean of branch-distance / euclidean-distance (path length ÷ straight-line distance) over branches with nonzero euclidean-distance; 1.0 = perfectly straight
branch_density	$n_branches / volume_um3 \times 10^6$ (branches per million μm^3 , i.e. per $\sim 100 \times 100 \times 100 \mu m$ cube — the $\times 10^6$ scaling exists purely to keep the numbers in a readable range)
endpoint_density	same normalisation, for $n_endpoints$
process_complexity	$n_endpoints \times mean_branch_len_um / volume_um3$ — a single combined index: more endpoints <i>and</i> longer branches <i>and</i> smaller cell body all push this up

A real gotcha, already found and fixed in this project once (see the branch-radius calibration work): `skan`'s `source_image=` parameter does **not** feed `summarize()`'s per-branch intensity/"mean-pixel-value" columns the way its name suggests — that requires baking the values into the skeleton array itself before constructing `skan.Skeleton`, not passing them as `source_image`. This module doesn't currently use any `source_image`-derived column (only geometry columns), so it isn't affected, but it's a real API trap worth knowing if this module is ever extended to pull intensity-along-skeleton statistics.

Failure mode: if `skan` isn't installed, or skeletonization produces an empty skeleton (can happen for a 1-2 voxel label), `_skeleton_stats` returns `(0, 0, 0.0, 0.0, 1.0)` rather than raising — all branch-derived columns read as zero/neutral for that label instead of aborting the batch.

6. Phase 2b — Intensity statistics (optional)

Only computed when `compute_stats()` is called with an image array (Tab 3's Image-layer selector). `_intensity_stats_worker` crops both the label mask and the source image to the label's bounding box, then:

```
mean_intensity      = mean(image[mask])
integrated_intensity = sum(image[mask])
intensity_cv        = std(image[mask]) / mean(image[mask])
(0.0 if mean == 0)
```

`intensity_cv` (coefficient of variation) is scale-independent — it measures relative spread of brightness within the cell, not absolute brightness, so it's comparable across cells or samples with different overall exposure/gain.

7. Post-assembly — Spatial statistics

`_spatial_stats(centroids_zyx_um, label_ids)` runs once on **all** centroids together (not per-label), using `scipy.spatial.cKDTree` for efficient nearest-neighbour queries — an $O(N \log N)$ k-d tree build rather than an $O(N^2)$ all-pairs distance matrix.

7.1 Nearest neighbours

```
tree = cKDTree(centroids)
dists, idxs = tree.query(centroids, k=3)  # self + 1st NN + 2nd
                                         NN
```

Column 0 of the result is always the point itself (distance 0), so the 1st nearest neighbour is column 1 and the 2nd is column 2.

`nearest_neighbor_label/nearest_neighbor_2_label` map the neighbour's *tree index* back to its actual label ID (tree order and label order aren't guaranteed to match). With exactly 2 cells total, `k` is clamped to 2 and the “2nd nearest neighbour” columns fall back to duplicating the 1st (there is no second neighbour to find).

7.2 Clark-Evans 3D index (`nearest_neighbor_ratio`)

The classical Clark & Evans (1954) index compares an observed mean nearest-neighbour distance to the value expected under **complete spatial randomness** (a homogeneous 3D Poisson point process) at the

same point density. This module computes a **per-cell** version — each cell’s own NND divided by the population’s expected NND — rather than the traditional single population-level summary statistic.

Derivation of the expected 3D nearest-neighbour distance, since this is worth having written down once: for a 3D Poisson process at intensity (density) ρ , the probability that no other point lies within radius r of a given point is the Poisson void probability for a sphere of volume $(4/3)\pi r^3$:

$$P(\text{NND} > r) = \exp(-\rho \cdot (4/3)\pi r^3)$$

The expectation of a nonnegative random variable is $E[X] = \int_0^\infty P(X > r) \, dr$, so:

$$E[\text{NND}] = \int_0^\infty \exp(-\rho \cdot (4/3)\pi \cdot r^3) \, dr$$

Substituting $u = r^3$ and using the standard Gamma-function integral $\int_0^\infty \exp(-a \cdot r^3) \, dr = (1/3) \cdot \Gamma(1/3) \cdot a^{(-1/3)}$, with $a = \rho(4/3)\pi$, and the identity $\Gamma(4/3) = (1/3)\Gamma(1/3)$:

$$E[\text{NND}] = \Gamma(4/3) \cdot (3 / (4\pi \cdot \rho))^{(1/3)}$$

— exactly what the code computes: `math.gamma(4/3) * (3.0 / (4.0*math.pi*density))**(1.0/3.0)`.

`nearest_neighbor_ratio = observed_NND / E[NND]`. Values **< 1** indicate the cell is closer to its neighbour than random chance would predict (local clustering); **> 1** indicates more spread out than random (regularity/dispersion).

A real approximation to know about: `density = N / bbox_volume`, where `bbox_volume` is the axis-aligned bounding box of the *centroid point cloud itself* (max - min per axis, multiplied together) — not the true tissue/brain volume the cells actually live in. If cells are sparse near the edges of their own bounding box (very likely for a roughly ellipsoidal brain region), this systematically **overestimates** density and therefore **underestimates** the expected NND, biasing `nearest_neighbor_ratio` slightly upward (toward “more regular than it really is”) near the population edges. Fine as a same-sample relative measure; treat with more caution when comparing the raw ratio across samples with different cell-count-derived bounding boxes.

7.3 local_density_100um

`tree.query_ball_point(centroids, r=100.0, return_length=True) - 1`
— count of other centroids within a 100 μm radius sphere of each cell (the -1 removes the point counting itself). A direct local crowding count, independent of the Clark-Evans model assumptions above.

7.4 depth_normalized

`clip(centroid_z_um / max(centroid_z_um), 0, 1)` across the current label set — 0 = the shallowest (lowest Z) cell in this result set, 1 = the deepest. Recomputed per-run, so it is **not** comparable across different fish/samples unless they share the same Z range by construction — it's a within-sample relative depth, not an absolute one.

7.5 is_volume_outlier (not computed by this module)

Unlike every other column on this page, `is_volume_outlier` is not produced inside `compute_stats()` at all — it's appended afterward, in `_widget.py`'s `_on_generate_stats()`, once the `DataFrame` comes back. It exists to hand the two “is this cell real?” edge cases GT sweeps and pipeline stages can't resolve on their own back to a human:

- **Too big** — could be two touching cells the pipeline's merge/split logic left joined instead of separating.
- **Too small** — could be genuine debris, or could be a real but unusually small microglia; below the deletion threshold, both routes already remove it outright (see [GUIDE.md §6a, Final min-size fraction](#) — Cellpose-SAM's final safety-net stage, the Pixel Classifier's own volume filter cutoff), but the gray zone between that threshold and the confirmed floor survives untouched on purpose on both routes.

```
too_big    = df["volume_vox"] > max_ceiling if max_ceiling is not
           None else False
too_small  = df["volume_vox"] < min_floor   if min_floor   is not
           None else False
df["is_volume_outlier"] = too_big | too_small
```

`max_ceiling` (`max_volume_recommended_vox` in config) and `min_floor` (`min_volume_recommended_vox`, the same field Common Settings' informative Min volume display reads) are both cross-fish histories tracked the same way as every Tab 5 sweep's recommendations — see `_update_gt_history()` in `_widget.py`. `min_floor` is a never-rising floor (`mode="min"`); `max_ceiling` is its never-falling mirror (`mode="max"`, the

same direction used for `branch_radius`). Both only move when Tab 3's **"This is verified ground truth"** checkbox is ticked for the run that measured them — an unverified/uncorrected prediction can widen neither bound, though every run, verified or not, is still flagged against whichever bounds were last confirmed.

8. Post-assembly — Brain region assignment (optional)

Only computed when Tab 3 is given a Shapes layer of boundary lines/polylines (drawn by the user, one curve per boundary between two anatomical regions) and a list of region names.

8.1 Geometric primitive: point-to-polyline projection

`_polyline_side_and_dist(cy, cx, pts)` finds, for a 2D point and a multi-segment polyline, which *segment* is nearest and which *side* of that segment the point falls on:

1. For every consecutive vertex pair (*a*, *b*) in the polyline, project the point onto the **infinite line** through *a*→*b*, then clamp the projection parameter *t* to $[0, 1]$ — this restricts the projection to the actual segment, not the infinite line (standard point-to-segment-distance technique).
2. Track the minimum distance across all segments — that segment is the "nearest segment."
3. At that nearest segment, compute the 2D cross product $dx_{seg} \cdot (cy - ay) - dy_{seg} \cdot (cx - ax)$. Its **sign** tells you which side of the segment's direction the point is on: this project's fish are oriented head (small X) → tail (large X), with boundary curves drawn top→bottom (increasing Y) by convention (see GUIDE.md §7a), so a negative cross product means the point is posterior to (right of) that boundary.

8.2 Region index from multiple boundaries

`_assign_brain_regions` sorts all supplied boundary curves by their **mean X coordinate** (ascending = most anterior first) — this makes boundary order well-defined regardless of the order the user drew them in. For each cell centroid, it counts how many boundaries the cell is posterior to:

```

idx = 0
for boundary in boundaries_sorted_anterior_to_posterior:
    if point_is_posterior_to(boundary):
        idx += 1
region = region_names[idx]

```

With k boundary curves this naturally produces $k+1$ regions (e.g. 1 boundary $\rightarrow$ anterior/posterior; 2 boundaries $\rightarrow$ 3 regions). `region_boundary_dist_um` is simply the minimum distance (in μm) from the cell's centroid to *any* boundary curve — cells near this value are close to a region edge and might reasonably be considered ambiguous/borderline.

9. Morphotype classification

`_classify_morphotype(sphericity, solidity, elongation, n_branches, surface_to_volume_ratio)` — a hand-tuned rule-based decision list, evaluated in priority order (first matching rule wins):

```

if elongation > 3.5 and n_branches <= 3:
    return "Rod-shaped"
if sphericity > 0.70 and solidity > 0.80 and n_branches <= 2:
    return "Amoeboid"
if n_branches >= 6 and sav > 2.0 and sphericity < 0.55:
    return "Ramified"
if n_branches >= 4 and sphericity < 0.65:
    return "Intermediate-ramified"
else:
    return "Intermediate"

```

This is **not** a fitted/learned classifier — it's a fixed set of thresholds chosen to match the standard microglia morphological categories used in the field (ramified/resting vs. amoeboid/activated, plus rod-shaped and intermediate states), evaluated directly on this module's own geometric outputs. Because it's rule-based and fully deterministic, it's reproducible and auditable, but it hasn't been validated against an independent ground-truth morphotype labeling — treat it as a descriptive heuristic, not a diagnostic classification, unless/until it's been checked against expert-annotated examples.

10. Natural-language description generation

The description column is produced by one of four interchangeable backends, selected via `backend_config["backend"]`:

Backend	Method	Requires
rule (default)	<code>_rule_based_description()</code> — a template that stitches together shape/surface/branch phrases from fixed thresholds on the already-computed columns (sphericity, solidity, elongation, branch/endpoint counts, morphotype)	nothing — fully offline, deterministic
ollama	POSTs a formatted prompt to a local Ollama server's <code>/api/generate</code> endpoint	a running local Ollama install, no API key
openai	POSTs to <code>{api_url}/v1/chat/completions</code> (OpenAI-compatible chat endpoint)	an API key
claude	POSTs to <code>https://api.anthropic.com/v1/messages</code>	an API key

The three API backends all format the same shared prompt template (`_STATS_PROMPT`) with that row's already-computed statistics (volume, elongation, dominant axis, sphericity, solidity, branch/endpoint counts, mean branch length, morphotype, centroid) and ask for **one sentence, max 40 words**. Every backend call is wrapped in a try/except that returns an inline `"[Backend error: ...]"` string on failure (timeout, bad key, unreachable server) rather than raising — one row's description failing doesn't abort the whole statistics run.

The rule-based backend is the only one exercised by anything else in the pipeline (it has no external dependency), and is what runs by default with no configuration.

11. Full column → algorithm cross-reference

Column	Section
label, volume_vox, volume_um3, centroid_*, bbox_*	§2
solidity, extent	§2
principal_axis_dir	§3.1
axis1_um, axis2_um, axis3_um	§3.2
elongation, eq_diam_um	§3.3
surface_area_um2, sphericity, surface_to_volume_ratio	§4
n_branches, n_endpoints, mean_branch_len_um, max_branch_len_um, branch_tortuosity, branch_density, endpoint_density, process_complexity	§5
mean_intensity, integrated_intensity, intensity_cv	§6
nearest_neighbor_label, nearest_neighbor_dist_um, nearest_neighbor_2_label, nearest_neighbor_2_dist_um	§7.1
nearest_neighbor_ratio	§7.2
local_density_100um	§7.3
depth_normalized	§7.4
is_volume_outlier	§7.5 — computed in <code>_widget.py</code> , not <code>_statistics.py</code>
brain_region, region_boundary_dist_um	§8
morphotype	§9
description	§10

For the plain-English meaning of every column (rather than the algorithm behind it), see [GUIDE.md §11](#).

12. References

- Wadell, H. (1935). *Volume, Shape, and Roundness of Quartz Particles*. Journal of Geology, 43(3) — the sphericity formula used in §4.
- Lorensen, W. E., & Cline, H. E. (1987). *Marching Cubes: A High Resolution 3D Surface Construction Algorithm*. ACM SIGGRAPH Computer Graphics, 21(4) — the surface-meshing algorithm behind `surface_area_um2` (§4).
- Clark, P. J., & Evans, F. C. (1954). *Distance to Nearest Neighbor as a Measure of Spatial Relationships in Populations*. Ecology, 35(4) — the spatial-randomness index generalised to 3D in §7.2.
- Nunez-Iglesias, J., et al. — `skan`: skeleton analysis in Python, used throughout §5.
- van der Walt, S., et al. (2014). *scikit-image: image processing in Python*. PeerJ 2:e453 — `regionprops_table`, `marching_cubes`, `skeletonize` (§2, §4, §5).
- Virtanen, P., et al. (2020). *SciPy 1.0*. Nature Methods — `cKDTree` (§7).